COMONADIC COHOMOLOGY AS AN EXTENSION OF GROUP COHOMOLOGY

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Abstract. In this article, we present group cohomology as a particular instance of comonadic cohomology on the category of G-sets. By constructing an adjunction between the category of G-sets and the category of $\mathbb{Z}[G]$-modules, we are able to compute this comonadic cohomology by means of the Ext functor. This approach not only provides a unified perspective on group cohomology within a comonadic framework, but also establishes a direct link with classical homological algebra. This connection reveals that group cohomology can be computed by varying the first entry of the Ext functor, thus providing a concrete computational approach via Ext over the category of $\mathbb{Z}[G]$-modules.

1. Introduction

2. Preliminaries

We will denote by Set the category of sets, by Ab the category of abelian groups, by $G-Set$ the category of G-sets for a fixed group G and by $R-Mod$ the category of R-modules for a fixed ring R (by ring we mean an associative unitary ring).

Definition 1. A **comonad** is an endofunctor F on some category $\mathcal{C}$ together with natural transformations $\varepsilon : F \to 1_{\mathcal{C}}$ and $\mu : F \to FF$, called respectively the **counit** and **comultiplication** of the comonad, such that the following diagrams commute.

$$\begin{tikzcd} F & FF \arrow[l, "F\varepsilon"'] \arrow[r, "\varepsilon F"] & F \\ & F \arrow[ul, equal] \arrow[u, "\mu"] \arrow[ur, equal] & \end{tikzcd} \qquad \begin{tikzcd} F \arrow[r, "\mu"] \arrow[d, "\mu"'] & FF \arrow[d, "F\mu"] \\ FF \arrow[r, "\mu F"] & FFF \end{tikzcd}$$

The first property is called **counitality** and the second **coassociativity**.

Definition 2 (Adjunction)**.** A pair of functors $\mathcal{C} \underset{L}{\overset{R}{\leftrightarrows}} \mathcal{D}$ is said to be an **adjoint pair** if there exist natural transformations $\eta : 1_{\mathcal{C}} \Rightarrow RL, \varepsilon : LR \Rightarrow 1_{\mathcal{D}}$ (called the **unit** and **counit** of the adjunction, respectively), such that the following two diagrams commute.

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$$\begin{tikzcd} L \arrow[r, "L\eta"] \arrow[dr, "1_L"'] & LRL \arrow[d, "\varepsilon L"] \\ & L \end{tikzcd} \qquad \begin{tikzcd} R \arrow[r, "\eta R"] \arrow[dr, "1_R"'] & RLR \arrow[d, "R\varepsilon"] \\ & R \end{tikzcd}$$

For a reference on comonads and adjunctions, we suggest [3].

The following is a well-known result which relates comonads to adjunctions.

Proposition 1. An adjoint pair $\mathcal{C} \underset{L}{\overset{R}{\leftrightarrows}} \mathcal{D}$ induces a comonad LR with counit ε and comultiplication $L\eta R$.

Definition 3. We denote by Δ the **simplex category**, which consists of:

- Objects of the form $[n] := \{0, ..., n\}$ for each $n \in \mathbb{N}$.
- Non-decreasing functions as morphisms.

Δ_{inj} will denote the subcategory of Δ containing only monomorphisms (injective functions $\delta_i^n : [n-1] \to [n]$ which skip the i-th term).

Definition 4. A **semisimplicial set** is a sequence of sets $\{X_n\}_{n\in\mathbb{N}}$ together with maps $\delta_i^n : X_{n+1} \to X_n$ (called **degeneracies**) for $i = 1, ..., n+1$ such that $\delta_i^n \circ \delta_j^{n+1} = \delta_{j-1}^n \circ \delta_i^{n+1}$ whenever $i < j$. The category of semisimplicial sets is denoted by $\mathcal{C}_{ss}$.

Remark. Equivalently, a semisimplicial set is a contravariant functor $X : \Delta_{inj}^{op} \to Set$. When X is a functor onto some category $\mathcal{C}$ instead of Set, it is called a **semisimplicial object**.

Proposition 2. Let (C, μ, ε) be a comonad on a category $\mathcal{C}$. Then $S_C(X)_n = C^n(X)$ together with $d_n^i = C^{n-i}\varepsilon C^i$ form a semisimplicial object for each $X \in \mathcal{C}$. Furthermore, this assignment is functorial.

Definition 5. A **chain complex** over a ring R is a family of R-modules $\{X_n\}_{n\in\mathbb{N}}$ together with morphisms $d_n : X_n \to X_{n-1}$ (called **differentials**) such that $d_n \circ d_{n+1} = 0$ for all $n \geq 0$. A morphism of chain complexes $\{(X_n, d_n)\}_{n\in\mathbb{N}} \to \{(Y_n, d'_n)\}_{n\in\mathbb{N}}$ is a family of R-module morphisms $\{\phi_n : X_n \to Y_n\}$ such that $\phi_{n-1} \circ d_n = d'_n \circ \phi_n$ (i.e. each square commutes).

$$\begin{tikzcd} \dots \arrow[r] & X_n \arrow[r, "d_n"] \arrow[d, "\phi_n"'] & X_{n-1} \arrow[r] \arrow[d, "\phi_{n-1}"] & \dots \\ \dots \arrow[r] & Y_n \arrow[r, "d'_n"'] & Y_{n-1} \arrow[r] & \dots \end{tikzcd}$$

Definition 6 (Group object)**.** A **group object** in a category $\mathcal{C}$ with terminal object 1 is an object G together with morphisms $m : G \times G \to G$, $e : 1 \to R$, $(_)^{-1} : G \to G$ satisfying the following axioms.

(1) m is associative. That is, the following diagram commutes.

$$\begin{tikzcd} G \times G \times G \arrow[r, "1_G \times m"] \arrow[d, "m \times 1_G"'] & G \times G \arrow[d, "m"] \\ G \times G \arrow[r, "m"'] & G \end{tikzcd}$$

(2) 1 acts as the neutral element:

$$\begin{tikzcd} G \arrow[r, "(e \circ p, 1_G)"] \arrow[d, "(1_G, e \circ p)"'] \arrow[dr, "1_G"] & G \times G \arrow[d, "m"] \\ G \times G \arrow[r, "m"'] & G \end{tikzcd}$$

where p is the unique morphism $G \to 1$.

(3) $(_)^{-1}$ acts as the inverse operation:

$$\begin{tikzcd} G \arrow[r, "((_)^{-1}, 1_G)"] \arrow[d, "(1_G, (_)^{-1})"'] \arrow[dr, "e \circ p"] & G \times G \arrow[d, "m"] \\ G \times G \arrow[r, "m"'] & G \end{tikzcd}$$

Furthermore, G is **abelian** if m is commutative; that is, $m\sigma = m$, where σ (called the *swap morphism*) is found via the universal property of $G \times G$ (i.e. $\sigma = \pi_2 \times \pi_1$).

Proposition 3. Let $G \in \mathcal{C}$. G is a group object if and only if the representable functor $Hom(_, G)$ factors through some $F : \mathcal{C} \to Grp$.

$$\begin{tikzcd} \mathcal{C} \arrow[r, "Hom(_,G)"] \arrow[d, "F"'] & Set \\ Grp \arrow[ur, hook] & \end{tikzcd}$$

Remark. Let G be a group.

- An abelian group object in $G-Set$ is a G-module, which is an abelian group together with a compatible G-action. It is easy to show that a G-module has a natural $\mathbb{Z}[G]$-module structura by defining the action $e_g \cdot x$ as $g \cdot x$ and extending linearly. Conversely, a $\mathbb{Z}[G]$-módule induces a G-module via the action $g \cdot x := e_g \cdot x$.
- In the case of $R - Mod$, the Hom set of any two modules is an abelian group, which implies that any R-module is an abelian group object in $R - Mod$.

Proposition 4. Let $\mathcal{C}$ be a category, G an abelian group object in $\mathcal{C}$ and $X \in \mathcal{C}_{ss}$ with degeneracies δ_i^n. We obtain a chain complex

$$Hom(X, G) : Hom(X_0, G) \xrightarrow{d_0} Hom(X_1, G) \xrightarrow{d_1} \dots$$

by taking $d_n = \sum_{i=0}^{n} (-1)^i Hom(\delta_i^n, G)$ for $n \geq 0$.

Proposition 5. Let $L \dashv R$ be an adjunction between categories $\mathcal{C}$ and $\mathcal{D}$. Recall that this induces a comonad $C = LR$ with counit ε. For each $X \in \mathcal{D}$ we obtain a semisimplicial set

$$S^C(X) : X \leftarrow C(X) \leftleftarrows C^2(X) \substack{\leftarrow \\ \leftarrow \\ \leftarrow} \dots$$

with degeneracies $(C^i \varepsilon C^{n-i})_X : C^{n+1}(X) \to C^n(X)$ for $n \geq 0$, $i = 0, ..., n$.

Furthermore, given a group G we obtain a chain complex

$$K^C(X, G) : Hom(S_0^C(X), G) \xrightarrow{d_0} Hom(S_1^C(X), G) \xrightarrow{d_1} \dots$$

by defining $d_n = \sum_{i=0}^{n} (-1)^i Hom((C^i \varepsilon C^{n-i})_X, G)$ for $n \geq 0$.

Definition 7. Let M be a G-module. Consider the group $Hom_{Set}(G^n, M)$ for each $n \neq 0$ ($G^0 := \{*\}$). Define $d^{n+1} : Hom_{Set}(G^n, M) \to Hom_{Set}(G^{n+1}, M)$ as

$$(d^{n+1} f)(g_1, ..., g_{n+1}) := g_1 f(g_2, ..., g_{n+1}) + \sum_{i=1}^{n} (-1)^i f(g_1, ..., g_i g_{i+1}, ..., g_{n+1}) + (-1)^{n+1} f(g_1, ..., g_n))$$

for $f : G^n \to M, g_1, ..., g_{n+1} \in G$. The n-th cohomology group of the cochain complex

$$Hom_{Set}(G^0, M) \xrightarrow{d_0} Hom_{Set}(G^1, M) \xrightarrow{d_1} \dots$$

is $H^n(G, M) := Ker(d^{n+1})/Im(d^n)$.

Remark. It is a well-known result

Definition 8. We define the n-th cohomology group of X with coefficients in G as $H^n_C(X, G) = H^n(Hom(K^C(X, G), G))$[5].

Definition 9. Let X be a set. Define $\mathbb{Z}^{(X)}$ to be the set of functions $X \to \mathbb{Z}$ with finite support.

Remark.

- For each $x \in X$, there is a function $e_x \in \mathbb{Z}^{(X)}$ such that $e_x(y) = \delta_{x,y}$ for any $y \in X$. Furthermore, $\{e_x : x \in X\}$ is a generating set for $\mathbb{Z}^{(X)}$.
- $\mathbb{Z}^{(X)}$ together with pointwise addition forms an abelian group.
- Taking $X = G$ a group, we can define $e_g e_h := e_{gh}$. Extending this product linearly to all of $\mathbb{Z}^{(G)}$ gives rise to an associative unitary ring. In this case we will use the notation $\mathbb{Z}[G]$ to differentiate it from the abelian group $\mathbb{Z}^{(G)}$.

3. Comonadic Cohomology

Proposition 6. Let R be a ring. The functors $Set \underset{F}{\overset{U}{\leftrightarrows}} R - Mod$ given by $F(X) = R^{(X)}$ (with the natural action) and U the forgetful functor, are adjoint.

Proof. The unit is $\eta_X(x) = e_X \in R^{(X)}$ for $X \in Set, x \in X$ and the counit is $\varepsilon_M(e_m) = m$ for $M \in R - Mod, m \in M$. We briefly verify the triangle identities:

$$(\varepsilon_{F(X)})(F\eta_X)(e_x) = (\varepsilon_{F(X)})(e_{e_x}) = e_x$$

$$(U\varepsilon_M)(\eta_{U(M)})(m) = (U\varepsilon_M)(e_m) = m$$

Thus, $F \dashv U$. ■

Proposition 7. Let G be a group. We have the following adjunctions.

(1) $F_1 : Set \to Ab$ given by $F_1(X) = \mathbb{Z}^{(X)}$ and $F_1(f : X \to Y)(e_x) = e_{f(x)}$ is the left adjoint of the forgetful functor $U_1 : Ab \to Set$.

(2) $F_2 : Ab \to \mathbb{Z}[G] - Mod$ given by $F_2(H) = \mathbb{Z}[G] \otimes_{\mathbb{Z}} H$ and $F_2(f : H \to K)(e_g \otimes_{\mathbb{Z}} h) = e_g \otimes_{\mathbb{Z}} f(h)$ is the left adjoint of the forgetful functor $U_2 : \mathbb{Z}[G] - Mod \to Ab$.

(3) $L_1 : Set \to G - Set$ given by $L_1(X) = G \times X$ (with the obvious group action) and $L_1(f : X \to Y)(g, x) = (g, f(x))$ is the left adjoint of the forgetful functor $R_1 : G - Set \to Set$.

(4) $L_2 : G - Set \to \mathbb{Z}[G] - Mod$ given by $L_2(X) = \mathbb{Z}^{(X)}$ and $L_2(f : X \to Y)(e_x) = e_{f(x)}$ is the left adjoint of the forgetful functor $R_2 : \mathbb{Z}[G] - Mod \to G - Set$, where $g \in G$ acts on $R_2(M) = M$ by $gm = e_g m$.

$$Set \xrightarrow{F_1} Ab \xrightarrow{F_2} \mathbb{Z}[G] - Mod$$
$$Set \xrightarrow{L_1} G - Set \xrightarrow{L_2} \mathbb{Z}[G] - Mod$$

Proof. The first three adjunctions are well-known. We briefly prove the last one by defining its unit and counit.

Take $\eta_X(x) = e_x$, $\varepsilon_M(\sum a_i e_{m_i}) = \sum a_i m_i$ for each $X \in G - Set$, $M \in \mathbb{Z}[G] - Mod$. Then

$$(\varepsilon_{L_2(X)})(L_2\eta_X)(e_x) = \varepsilon_{L_2(X)}(e_{e_x}) = e_x$$

On the other hand,

$$(R_2\varepsilon_M)(\eta_{R_2(M)}(m)) = (R_2\varepsilon_M)(e_m) = m$$

Thus, $L_2 \dashv R_2$.

■

Proposition 8. Let G be a group. The following diagram commutes.

$$\begin{tikzcd} Set \arrow[r, "F_1"] \arrow[d, "L_1"'] & Ab \arrow[d, "F_2"] \\ G - Set \arrow[r, "L_2"'] & \mathbb{Z}[G] - Mod \end{tikzcd}$$

Proof. Let X be a set. On one hand, $L_2L_1(X) = L_2(G \times X) = \mathbb{Z}^{(G\times X)}$. On the other, $F_2F_1(X) = F_2(\mathbb{Z}^{(X)}) = \mathbb{Z}[G] \otimes_{\mathbb{Z}} \mathbb{Z}^{(X)} \cong (\mathbb{Z}[G] \otimes_{\mathbb{Z}} \mathbb{Z})^{(X)} \cong \mathbb{Z}[G]^{(X)}$. We must show, then, that $\mathbb{Z}^{(G\times X)} \cong \mathbb{Z}[G]^{(X)}$ as $\mathbb{Z}[G]$-modules.

Recall that $\mathbb{Z}^{(G\times X)} = \{s : G \times X \to \mathbb{Z} : s \text{ has finite support}\}$ and $\mathbb{Z}[G]^{(X)} = \{t : X \to \mathbb{Z}[G] : t \text{ has finite support}\}$. Define $\phi_X : \mathbb{Z}^{(G\times X)} \to \mathbb{Z}[G]^{(X)}$ by $\phi_X(s)(x) : g \to s(g, x)$. Note that if $s \in \mathbb{Z}^{(G\times X)}$ has support $supp(s) = \{(g_1, x_1), ..., (g_n, x_n)\}$, then $supp(\phi_X(s)(x)) \subseteq \{g_1, ..., g_n\}$ and $supp(\phi_X(s)) = \{x_!, ..., x_n\}$. This implies that $\phi_X(s)(x) \in \mathbb{Z}[G]$ and $\phi_X(s) \in \mathbb{Z}[X]^{(X)}$.

Let us verify that ϕ_X is a $Z[G] - Mod$ morphism. If $s_1, s_2 \in \mathbb{Z}^{(G\times X)}$ and $u \in \mathbb{Z}[G]$, then

$$\phi_X(us_1 + s_2)(x)(g) = (us_1 + s_2)(g, x) = u(g)s_1(g, x) + s_2(g, x) = (u\phi_X(s_1) + \phi_X(s_2))(x)(g)$$

This means that $\phi_X(us_1 + s_2) = u\phi_X(s_1) + \phi_X(s_2)$, as desired.

Next, we check that ϕ_X is bijective. If $\phi_X(s) = 0$, then $\phi_X(s)(x) = 0$ for all $x \in X$. In other words, $s(g, x) = 0$ for all $x \in X$, $g \in G$, and so $s = 0$. Thus, ϕ_X is injective. Now, if $t \in \mathbb{Z}[G]^{(X)}$, take $s_t : G\times X \to \mathbb{Z}$

given by $(g,x) \mapsto t(x)(g)$. If $supp(t) = \{x_1, ..., x_n\}$ and $supp(t(x_i)) = \{g_{i_1}, ..., g_{i_n}\}$ for each x_i (we may assume without loss of generality that $supp(t)$ and each $supp(t(x_i))$ all have the same cardinality), then $supp(s_t) = \cup_{i=1}^{n} supp(t(x_i)) \times \{x_i\}$. That is, s_t has finite support and so $s_t \in \mathbb{Z}^{(G\times X)}$. It is also clear that $\phi_X(s_t) = t$.

Lastly, let us check that ϕ is natural. If $f : X \to Y$ is a function, we obtain the following diagram.

$$\begin{tikzcd} \mathbb{Z}^{(G\times X)} \arrow[r, "\phi_X"] \arrow[d, "L_2L_1(f)"'] & \mathbb{Z}[G]^{(X)} \arrow[d, "F_2F_1(f)"] \\ \mathbb{Z}^{(G\times Y)} \arrow[r, "\phi_Y"'] & \mathbb{Z}[G]^{(Y)} \end{tikzcd}$$

Given $x \in X$ and $g \in G$, evaluating the top side of the diagram on $e_{g,x}$ gives

$$F_2F_1(f)\phi_X(e_{g,x}) = F_2F_1(f)(d_{x,g}) = d_{f(x),g}$$

while the bottom side gives

$$\phi_Y L_2L_1(f)(e_{g,x}) = \phi_Y(e_{g,f(x)}) = d_{f(x),g}$$

Thus, ϕ is natural. This concludes the proof. ■

Each adjunction in the previous proposition induces a comonad. Throughout the rest of the paper, we will denote $T = FU$, $C = LR$, $T_i = F_iU_i$ and $C_i = L_iR_i$ (for $i = 1, 2$). Note that $T \simeq C$.

Proposition 9. Let $X \in G - Set$. Then $L_2(S^{C_1}(X)) \cong S^T(L_2(X))$.

Proof. Let $\varepsilon_1, \varepsilon$ bet the counits of C_1 and T their respective counits. Note that in the semisimplicial set

$$S^T(L_2(X)) : L_2(X) \leftarrow \mathbb{Z}^{(G\times L_2(X))} \leftleftarrows \mathbb{Z}^{(G\times \mathbb{Z}^{(G\times L_2(X))})} \substack{\leftarrow\\\leftarrow\\\leftarrow} ...$$

we have $\mathbb{Z}^{(G\times \mathbb{Z}^{(G\times L_2(X))})} \cong \mathbb{Z}^{(G^2\times L_2(X))}$ (the isomorphism is given by assigning each basis element $e_{(g_1,e_{(g_2,e_x)})}$ to $e_{(g_1,g_2,e_x)}$ and vice versa). In general, the n-th term of $S^T(L_2(X))$ is isomorphic to $\mathbb{Z}^{(G^n\times L_2(X))}$ for $n > 0$.

Thus, the semisimplicial sets in question are

$$L_2(S^{C_1}(X)) : \mathbb{Z}^{(X)} \leftarrow \mathbb{Z}^{(G\times X)} \leftleftarrows \mathbb{Z}^{(G^2\times X)} \substack{\leftarrow\\\leftarrow\\\leftarrow} ...$$

and

$$S^T(L_2(X)) : \mathbb{Z}^{(X)} \leftarrow \mathbb{Z}^{(G\times \mathbb{Z}^{(X)})} \leftleftarrows \mathbb{Z}^{(G^2\times \mathbb{Z}^{(X)})} \substack{\leftarrow\\\leftarrow\\\leftarrow} ...$$

We define $\phi_n : \mathbb{Z}^{(G^n\times X)} \to \mathbb{Z}^{(G^n\times \mathbb{Z}^{(X)})}$ for $n \geq 1$ (ϕ_0 is the identity) by mapping $e_{(g_1,...,g_n,x)}$ to $e_{(g_1,...,g_n,e_x)}$. Let us check that this family is an isomorphism of chain complexes.

$$\begin{tikzcd}
\mathbb{Z}^{(X)} \arrow[d, "\phi_0"'] & \mathbb{Z}^{(G\times X)} \arrow[l, "L_2(\varepsilon_1)_X"] \arrow[d, "\phi_1"'] & \mathbb{Z}^{(G^2\times X)} \arrow[l, "L_2(C_1^i\varepsilon_1 C_1^{1-i})_X"] \arrow[d, "\phi_2"'] & \cdots \arrow[l, "L_2(C_1^i\varepsilon_1 C_1^{2-i})_X"] \\
\mathbb{Z}^{(X)} & \mathbb{Z}^{(G\times\mathbb{Z}^{(X)})} \arrow[l, "\varepsilon_X"] & \mathbb{Z}^{(G^2\times\mathbb{Z}^{(X)})} \arrow[l, "(T^i\varepsilon T^{1-i})_{L_2(X)}"] & \cdots \arrow[l, "(T^i\varepsilon T^{2-i})_{L_2(X)}"]
\end{tikzcd}$$

Claim. Each ϕ_n is an isomorphism of $\mathbb{Z}[G]$-modules.
The case $n = 0$ is clear. For $n > 0$, ϕ_n defines a bijection between the bases $\{e_{(g_1,\ldots,g_n,x)}\}_{g_i\in G, x\in X}$ and $\{e_{(g_1,\ldots,g_n,e_x)}\}_{g_i\in G, x\in X}$, so it suffices to check that it is a $\mathbb{Z}[G]$-module morphism. This is simply done by recalling the action of $\mathbb{Z}[G]$ on $\mathbb{Z}^{(G^n\times X)}$ and $\mathbb{Z}^{(G^n\times\mathbb{Z}^{(X)})}$:

$$\phi_n(h\cdot e_{(g_1,\ldots,g_n,x)}) = \phi_n(e_{(hg_1,\ldots,g_n,x)}) = e_{(hg_1,\ldots,g_n,e_x)} = h\cdot\phi_n(e_{(g_1,\ldots,g_n,x)})$$

for all $h \in G$. Thus, $\mathbb{Z}^{(G^n\times X)} \overset{\phi_n}{\cong} \mathbb{Z}^{(G^n\times\mathbb{Z}^{(X)})}$.

Claim. The family $\{\phi_n\}_{n\in\mathbb{N}}$ is a morphism of semisimplicial sets.
Given a basis element $e_{(g_1,\ldots,g_{n+1},x)} \in \mathbb{Z}^{(G^{n+1}\times X)}$, we have

$$\begin{aligned}(\phi_n\circ L_2(C_1^i\varepsilon_1 C_1^{n-i})_X)(e_{(g_1,\ldots,g_{n+1},x)}) &= \phi_n(e_{(g_1,\ldots,g_{i+1}g_{i+2},\ldots,g_{n+1},x)})\\ &= e_{(g_1,\ldots,g_{i+1}g_{i+2},\ldots,g_{n+1},e_x)}\end{aligned}$$

$$\begin{aligned}((T^i\varepsilon T^{n-i})_{L_2(X)}\circ\phi_{n+1})(e_{(g_1,\ldots,g_{n+1},x)}) &= (T^i\varepsilon T^{n-i})_{L_2(X)}(e_{(g_1,\ldots,g_{n+1},e_x)})\\ &= e_{(g_1,\ldots,g_{i+1}g_{i+2},\ldots,g_{n+1},e_x)}\end{aligned}$$

Extending linearly, we conclude that $\phi_n\circ L_2(C_1^i\varepsilon_1 C_1^{n-i})_X = (T^i\varepsilon T^{n-i})_{L_2(X)}\circ\phi_{n+1}$ for all i, as desired. ■

Corollary 1. Let X be a G-set. Then $K^{C_1}(X,G) \cong K^T(L_2(X),G)$. Consequently, $H^n_{C_1}(X,G) \cong H^n_T(L_2(X),G)$ for all n.

Proposition 10. Let X be a G-Set and M be a $\mathbb{Z}[G]$-module. The following chain complexes are isomorphic:

$$K^{C_1}(X,M) : Hom_G(X,M) \xrightarrow{d_0'} Hom_G(G\times X,M) \xrightarrow{d_1'} Hom_G(G^2\times X,M) \xrightarrow{d_2'} \ldots$$

$$K^C(L_2(X),M) : Hom_{\mathbb{Z}[G]}(\mathbb{Z}^{(X)},M) \xrightarrow{f_0'} Hom_{\mathbb{Z}[G]}(\mathbb{Z}^{(G\times X)},M) \xrightarrow{f_1'} Hom_{\mathbb{Z}[G]}(\mathbb{Z}^{(G^2\times X)},M) \xrightarrow{f_2'} \ldots$$

where $d_n' = Hom_G(_,M)(d_0)$ and $f_n' = Hom_{\mathbb{Z}[G]}(_,M)(L_2(d_n))$.

Proof. Let $\alpha \in Hom_G(G^n\times X,M), \beta \in Hom_{\mathbb{Z}[G]}(\mathbb{Z}^{(G^n\times X)},M)$. Consider $\phi_n : Hom_G(G^k\times X,M) \to Hom_{\mathbb{Z}[G]}(\mathbb{Z}^{(G^k\times X)},M)$ and
$\psi_n : Hom_{\mathbb{Z}[G]}(\mathbb{Z}^{(G^k\times X)},M) \to Hom_G(G^k\times X,M)$ defined by

$$\phi_n(\alpha)(e_{(g_1,\ldots,g_n,x)}) = \alpha(g_1,\ldots,g_n,x)$$

$$\psi_n(\beta)(g_1,\ldots,g_n,x) = \beta(e_{(g_1,\ldots,g_n,x)})$$

Claim. ϕ_n is a well defined homomorphism for all $n \geq 0$.
Given $f \in Hom_G(G^n \times X, M)$ and $h \in G$, we have

$$\begin{aligned}\phi_n(\alpha)(e_h \cdot e_{(g_1,\ldots,g_n,x)}) &= \phi_n(\alpha)(e_{(hg_1,\ldots,g_n,x)}) \\ &= \alpha(hg_1, ..., g_n, x) \\ &= h \cdot \alpha(g_1, ..., g_n, x) \\ &= h \cdot \phi_k(\alpha)(e_{(g_1,\ldots,g_n,x)})\end{aligned}$$

so ϕ_n is well defined. Additionally,

$$\begin{aligned}\phi_n(\alpha_1 + \alpha_2)(e_{(g_1,\ldots,g_n,x)}) &= (\alpha_1 + \alpha_2)(g_1, ..., g_n, x) \\ &= \alpha_1(g_1, ..., g_n, x) + \alpha_2(g_1, ..., g_n, x) \\ &= \phi_n(\alpha_1(e_{(g_1,\ldots,g_n,x)}) + \phi_n(\alpha_2(e_{(g_1,\ldots,g_n,x)})\end{aligned}$$

for $\alpha_1, \alpha_2 \in Hom_G(G^n \times X, M)$.

A similar reasoning shows that each ψ_n is a well defined homomorphism.

Next, we show that $\{\phi_n\}$ is a morphism of chain complexes.

$$\begin{tikzcd}
Hom_G(X, M) \arrow[r, "d'_0"] \arrow[d, "\phi_0"'] & Hom_G(G \times X, M) \arrow[r, "d'_1"] \arrow[d, "\phi_1"'] & Hom_G(G^2 \times X, M) \arrow[r, "d'_2"] \arrow[d, "\phi_2"'] & \ldots \\
Hom_{\mathbb{Z}[G]}(\mathbb{Z}^{(X)}, M) \arrow[r, "f'_0"'] & Hom_{\mathbb{Z}[G]}(\mathbb{Z}^{(G\times X)}, M) \arrow[r, "f'_1"'] & Hom_{\mathbb{Z}[G]}(\mathbb{Z}^{(G^2\times X)}, M) \arrow[r, "f'_2"'] & \ldots
\end{tikzcd}$$

For arbitrary n, composing along the top side gives

$$(\phi_{n+1} \circ d'_n)(\alpha)(e_{(g_1,,,,g_n,x)}) = d'_0(\alpha)(g_1, ..., g_n, x) = \alpha(g_1g_2, ..., g_n, x)$$

whereas the bottom side gives

$$(f'_n \circ \phi_n)(\alpha)(e_{(g_1,,,,g_n,x)}) = \phi_n(\alpha)(e_{(g_1g_2,\ldots,g_n,x)}) = \alpha(g_1g_2, ..., g_n, x)$$

so each square of the above diagram commutes. A similar argument shows $\{\psi_n\}$ is also a morphism of chain complexes.

Lastly, we check that $\psi_n = \phi_n^{-1}$ for each $n \geq 0$:

$$(\phi_n \circ \psi_n)(\alpha)(e_{(g_1,,,.g,_n,x)}) = \phi_n(\alpha)(g_1, ..., g_n, x) = \alpha(g_1, ..., g_n, x)$$

$$(\psi_n \circ \phi_n)(\beta)(g_1, ..., g_n, x) = \phi_n(\beta)(e_{(g_1,,,.g,_n,x)}) = \beta(g_1, ..., g_n, x)$$

This concludes the proof. ■

4. Results

Let G be a group, X a G-set and M a G-module. M can also be seen as a $\mathbb{Z}[G]$-module and so $H^n_C(X,M) \cong Ext^n_{\mathbb{Z}[G]}(Z^{(X)},M)$[<**empty citation**>] (Weibel?). Combining this with corollary 1, we have

$$H^n_{C_1}(X,M) \cong H^n_C(\mathbb{Z}^{(X)},M) \cong H^n_T(\mathbb{Z}^{(X)},M) \cong Ext^n_{\mathbb{Z}[G]}(\mathbb{Z}^{(X)},M)$$

Thus, $H^n_{C_1}(X,M) \cong Ext^n_{\mathbb{Z}[G]}(\mathbb{Z}^{(X)},M)$. In particular, if X consists of a single point then

$$H^n_{C_1}(*,M) \cong Ext^n_{\mathbb{Z}[G]}(\mathbb{Z},M)$$

It is well known that the classical group cohomology $H^n(G,M)$ is naturally isomorphic to $H^n_{C_1}(*,M)$[1], which means that $H^n(G,M) \cong Ext^n_{\mathbb{Z}[G]}(\mathbb{Z},M)$. This is a known result, but the comonadic cohomology provides an alternative proof to the classical cohomological calculations.

$\mathbb{Z}[G]$ cubre a $\mathbb{Z}$ tomando $e_g \mapsto 1$. El núcleo es $e_g - e_h$. ¿Hay otra forma de escribirlo?

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